

# NeoAI Tech Internship Selection Assessment

## Data Analytics Report

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**Tools Used:** Google Colab, Python, Pandas, Matplotlib

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### 1. Executive Summary

This report presents the analysis performed on the healthcare dataset provided as part of the NeoAI Tech Data Analytics Internship Selection Assessment. The objective was to clean the dataset, perform exploratory data analysis, and generate meaningful business and demographic insights using Python.

The analysis focused on identifying treatment cost trends, department-wise revenue, insurance coverage, demographic patterns, and treatment outcomes. Data preprocessing techniques were applied to ensure data quality before conducting the analysis.

The complete implementation, including Python code and visualizations, is available in the accompanying Jupyter Notebook.

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### 2. Dataset Overview

The provided healthcare dataset consists of patient records containing demographic information, disease details, hospital departments, treatment costs, insurance status, blood groups, treatment outcomes, and city information.

The original dataset contained **515 healthcare records**. During preprocessing, **15 duplicate records** were identified and removed, resulting in a final dataset of **500 unique records** for analysis.

The primary objective of the analysis was to answer the business and patient demographic questions provided in the assessment while identifying meaningful trends that could support healthcare decision-making.

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### 3. Data Cleaning & Preprocessing

Before performing the analysis, the dataset was carefully examined to ensure data quality and consistency.

The following preprocessing steps were performed:

- Imported the dataset into a Pandas DataFrame.
- Verified dataset dimensions and column names.
- Checked data types for all attributes.
- Checked for missing values.
- Identified duplicate records.
- Removed 15 duplicate records.
- Generated descriptive statistics for numerical attributes.
- Confirmed the dataset was suitable for further analysis.

These preprocessing steps ensured that the analysis was performed on accurate and reliable data.

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## 4. Business Analysis

### Question 1

**Which disease contributes the highest total treatment cost?**

**Result**

**Disease:** Asthma

**Total Treatment Cost:** ₹7,930,109

**Insight**

Asthma accounts for the highest cumulative treatment expenditure in the dataset. This indicates that managing asthma requires significant healthcare resources and contributes substantially to overall treatment costs.

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### Question 2

**Which department generates the highest revenue?**

**Result**

**Department:** Neurology

**Revenue Generated:** ₹11,581,883

### **Insight**

Neurology generated the highest revenue among all departments. This may indicate higher patient volume, specialized treatments, or relatively higher treatment costs within the department.

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### **Question 3**

**What is the average treatment cost by disease and insurance status?**

#### **Result**

Average treatment costs were calculated for every disease by comparing insured and uninsured patients.

#### **Insight**

The comparison shows noticeable variations in treatment costs across diseases and insurance categories, highlighting the financial impact of insurance coverage on healthcare expenditure.

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### **Question 4**

**What percentage of total treatment costs are covered by insured patients?**

#### **Result**

**Insurance Coverage: 51.22%**

#### **Insight**

More than half of the total treatment expenditure is covered by insured patients. This demonstrates the significant role insurance plays in reducing the financial burden on patients.

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### **Question 5**

**Which city has the highest average treatment cost?**

#### **Result**

**City: Delhi**

**Average Treatment Cost: ₹138,971.61**

#### **Insight**

Delhi recorded the highest average treatment cost per patient, indicating relatively higher healthcare expenses compared to other cities in the dataset.

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## 5. Patient Demographics Analysis

### Question 6

Which age group contributes the highest healthcare expenditure?

**Result**

**Age Group:** 60+

**Total Expenditure:** ₹19,229,766

**Insight**

Patients aged 60 years and above contribute the highest healthcare expenditure, reflecting the increased healthcare requirements associated with older age groups.

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### Question 7

What diseases are most common among different age groups?

**Age Group Most Common Disease**

|       |               |
|-------|---------------|
| 18–30 | Asthma        |
| 31–45 | Heart Disease |
| 46–60 | Arthritis     |
| 60+   | Pneumonia     |

**Insight**

Disease prevalence varies significantly across different age groups. These trends can help healthcare providers focus preventive measures and allocate medical resources more effectively.

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### Question 8

Is there a significant difference in treatment costs between male and female patients?

## Result

Average Treatment Cost (Female): ₹123,070.07

Average Treatment Cost (Male): ₹119,904.77

## Insight

Female patients have a slightly higher average treatment cost than male patients in this dataset. However, the difference is relatively small.

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## Question 9

**Which blood group has the highest incidence of critical outcomes?**

## Result

**Blood Group:** AB–

**Critical Outcome Rate:** 48.08%

## Insight

Among all blood groups, AB– recorded the highest proportion of critical treatment outcomes in the dataset.

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## 6. Key Findings

The analysis identified several important trends within the healthcare dataset:

- Asthma contributes the highest total treatment expenditure.
  - Neurology generates the highest hospital revenue.
  - Insurance covers **51.22%** of the total treatment costs.
  - Delhi records the highest average treatment cost per patient.
  - Patients aged **60+** account for the highest healthcare expenditure.
  - Disease prevalence varies across different age groups.
  - Female patients have a slightly higher average treatment cost than male patients.
  - AB– blood group shows the highest incidence of critical outcomes.
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## **7. Conclusion**

This assessment demonstrates the practical application of Python-based data analytics techniques for healthcare data analysis. The dataset was successfully cleaned, explored, and analysed to answer all the required business and patient demographic questions.

The analysis provides valuable insights into treatment expenditure, departmental revenue, insurance coverage, patient demographics, and healthcare outcomes. These findings can support better resource allocation, financial planning, and informed decision-making within healthcare organizations.

The accompanying Jupyter Notebook contains the complete implementation, data preprocessing steps, analysis, and visualizations used to obtain these results.